

# Daniel Alfredo Barreras Meraz

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## EXPERIENCE

### Research Engineer

Aug. 2025 – Present

*Harvard Medical School - Boston Children's Hospital*

*Boston, MA*

- Engineered production-scale Python pipeline processing distributed datasets across 6 global medical sites, implementing automated ETL workflows with Docker containerization for 10,000+ volumetric medical samples.
- Built scalable data processing infrastructure reducing batch processing time 75% (3-5 hours to 30-60 minutes) through parallel processing, caching strategies, and workflow optimization across distributed systems.
- Orchestrated 8-stage automated workflow for multi-site data acquisition, validation, and processing under strict regulatory compliance (HIPAA, IRB protocols), ensuring data integrity and audit trails for clinical research.

### Sr Full Stack Engineer

Nov. 2023 – Present

*Sidepocket Inc*

*San Francisco, CA (Remote)*

- Spearheaded architecture and mentored 3 engineers while scaling microservices from 2K to 15K users via Flask/Spring Boot design, maintaining 99.9% uptime and cutting infrastructure costs 30%.
- Delivered high-throughput RESTful APIs handling 2M+ daily transactions with sub-200ms latency via PostgreSQL tuning, Redis caching, and distributed system patterns.
- Transformed CI/CD deployment on AWS ECS, eliminating 85% manual overhead through infrastructure-as-code practices enabling 5x faster releases.

### Database Administrator

Jun. 2024 – Aug. 2025

*Ciena Corp*

*Zapopan, MX*

- Developed Python-based workflow automation with AWS CloudWatch integration, eliminating 70% manual operations while processing 10M+ daily records across Oracle EBS infrastructure.
- Accelerated mission-critical PL/SQL queries 45% (2.1s to 800ms response) by applying indexing strategies, execution plan analysis, and query refactoring across distributed databases.

### Full Stack Developer

Feb. 2021 – Nov. 2023

*Molina Group*

*Hermosillo, MX*

- Launched Spring Boot microservices supporting 1,200+ geospatially-distributed vineyard locations, generating \$200K+ annual revenue increase through 30% operational efficiency gains via workflow automation.
- Built Next.js frontend with optimized React components and visualization libraries, boosting geospatial rendering performance 50% and improving user engagement 40%.
- Secured enterprise authentication infrastructure using OAuth2 and AWS Cognito for 600+ users, achieving SOC 2 compliance and establishing role-based access control.

## RESEARCH & ML PROJECTS

### Phoenix Eye: AI Wildfire Detection | *Python, TensorFlow, React.js, Azure*

Sep. 2023 – Dec. 2024

- Delivered geospatial ML application achieving 95% detection accuracy across 50,000+ satellite images. NASA Space Apps Global Finalist (Top 40/5,000+ teams) and Dubai Prototypes for Humanity 2024 Finalist.

## EDUCATION

### Tecnológico de Monterrey (ITESM)

Guadalajara, MX

*B.S. in Computer Science and Technology, GPA: 98/100*

## TECHNICAL SKILLS

**Languages:** Python, Java, JavaScript/TypeScript, C++, Go, PL/SQL, MATLAB, R, Scala

**ML & AI:** PyTorch, TensorFlow, Computer Vision, Deep Learning, Geospatial Analysis, NLP, scikit-learn

**Backend:** Spring Boot, Flask, FastAPI, Microservices, Distributed Systems, RESTful APIs, GraphQL

**Cloud & DevOps:** AWS (ECS, Lambda, RDS, S3, CloudWatch, EC2), Docker, Kubernetes, CI/CD, Terraform

**Databases:** PostgreSQL, MongoDB, Oracle, DynamoDB, Redis, Workflow Automation, ETL Pipelines

**Frontend:** React, Next.js, Vue.js, D3.js, Geospatial Visualization, HTML5/CSS3